[INCH-POUND]

MIL-C-83517
AMENDMENT 3
15 April 1994
SUPERSEDING
AMENDMENT 2
9 December 1993

MILITARY SPECIFICATION

CONNECTOR, COAXIAL, RADIO FREQUENCY FOR COAXIAL,
STRIP OR MICROSTRIP TRANSMISSION LINE

GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-C-83517, dated 20 September 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2 1.1, standards, military, following MIL-STD-202, add the following:

"MIL-STD-790 - Reliability Assurance Program for Electronic Parts Specifications."

PAGE 3

2.2, following the American Society for Testing and Materials address, add the following:

"ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA 557 - EIA Standard Implementation of Statistical Process Control (SPC) in Manufacturing Process.

(Application for copies should be addressed to the Electronic Industries Association, Engineering Office, 2001 Eye Street, N.W., Washington, DC 20006.)"

Add the following paragraph:

"3.2 1 Product assurance requirements. The product assurance requirements of the connector sources furnished under this specification shall be established and maintained in accordance with the procedures and requirements specified in MIL-STD-790 with details specified to 4.1.3."

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3 23, delete and substitute:

"3.23 Marking. Connectors and associated fittings shall be permanently and legibly marked in accordance with the general marking requirements of MIL-STD-130 with the military part number (see 1.2.1), manufacturer's federal supply code, and final assembly date code The marking location is optional. When practicable, a location should be picked that will least likely be covered in cable assembly or installation "

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Add the following paragraph

"3 23 1 JAN and J marking The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number 1/ Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or associated detail specifications, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN"."

The "JAN" or "J" is not part of the part number but indicates a certification "

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Section 4, delete paragraphs 4 through 4 5 2 and substitute paragraphs 4 through 4.5.3 as follows.

"4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4 1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.3 Product assurance program. A product assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows.

a Qualification inspection (see 4 4)

b Quality conformance inspection (see 4 5)

4 3 Inspection conditions Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202 and MIL-STD-1344 For each test of threaded coupling connectors, where the test is performed on mated pairs, the pair shall be torqued to the specified value (see 3 1)

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4 4 Qualification inspection Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6 4) on sample units produced with equipment and procedures normally used in production

4 4 1 Sample size. Nine connectors of the same part number with its mating connector shall be subjected to qualification inspection

4 4.2 Group qualification For group qualification of all series of connectors covered by this specification, see 3.1 The Government reserves the right to authorize performance of any or all qualification inspection of additional types in the group that are considered necessary for qualification within each group.

4 4 3 Inspection routine The sample shall be subjected to the inspections specified in table II All sample units shall be subjected to the inspection of group I. The sample units shall then be divided into three groups consisting of three connectors. The sample units shall then be subjected to the inspection for their particular group and in the sequence given for that group.

4.4.4 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.4.5 Retention of qualification To retain qualification, the contractor shall forward a report at 12- or 36-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date Initial retention of qualification shall be at a 12-month interval; subsequent retention of qualification at a 36-month interval The report shall consist of:

a. A summary of the results of the test performed for inspection of product for delivery, groups A and B, indicating as a minimum the number of lots that have passed and the number that have failed shall be submitted every 12 months. The results of tests of all reworked lots shall be identified and accounted for.

b. A summary of the results of test performed for periodic inspection, group C, including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 36-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 60 days after the end of each 12- or 36-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 12- or 36-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

NOTE: In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during 3 consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product from each group, as defined by 4.4.2 to testing in accordance with the qualification inspection requirements.

4.4.6 Extension of qualification. Manufacturers who have products listed on QPL-39012 and produce connectors of the same series in accordance with this specification, may apply to the qualifying activity for extension of qualification to this specification, provided the interfacial coupling, materials, and plating of the connectors are identical, and the connector successfully meets the requirements of groups I, II, and III of table II.

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TABLE II Qualification inspection

Inspection	Requirement paragraph	Test method paragraph
Group I		
Visual and mechanical inspection		
Material	3 3	4.6.1
Finish	3 3 1	4 6 1
Dissimilar metals	3 3.2	4.6.1
Design and construction (dimensions)	3.4	4.6.1
Marking	3.23	4.6.1
Force to engage/disengage	3.5	4.6.2
Coupling force torque	3.6	4.6.3
Design and construction	3.4	4.6.1
Force to engage/disengage	3.5	4.6.2
Contact gaging	3.7	4.6.4
Permeability of nonmagnetic materials	3.8	4.6.5
Workmanship	3.24	4.6.1
Seal	3 9	4.6.6
Hermetic sealed connectors	3.9.1	4.6.6 1
Pressurized and waterproofed connectors	3.9.2	4.6.6 2
Insulation resistance	3.10	4 6.7
Group II 1/		
Center contact retention	3.11	4.6.8
Axial force	3.11.1	4.6.8 1
Torque	3.11.2	4.6.8.2
Dielectric withstanding voltage	3.12	4.6.9
Corrosion	3.13	4.6.10
Force to engage/disengage	3.5	4.6.2
Group III		
VSWR	3.14	4.6.11
RF transmission loss 2/	3.15	4.6.12
RF leakage 2/	3.16	4.6.13
Connector durability	3 17	4.6.14
Force to engage/disengage	3.5	4.6.2
Contact gaging	3.7	4.6 4
Group IV		
Contact resistance	3.18	4.6.15
Thermal shock	3.19	4.6.16
Dielectric withstanding voltage	3.12	4.6.9
Contact resistance	3.18	4.6.15
VSWR	3.14	4.6.11
Moisture resistance	3 20	4.6.17
Dielectric withstanding voltage	3.12	4.6.9
Insulation resistance	3.10	4 6 7
RF high potential withstanding voltage 2/	3.21	4 6 18
Coupling mechanism retention force	3.22	4 6 19
Force to engage/disengage	3.5	4 6 2

1/ See 3 1
2/ These tests to be performed only during initial qualification as long as the qualifying design and manufacturing process has not been changed

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4.5 Quality conformance inspection

4.5 1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection

4 5 1.1 Inspection lot. An inspection lot shall consist of all connectors of the same part number produced under essentially the same conditions, and offered for inspection at one time

4 5 1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table III in the order shown

4.5 1.2.1 Sampling plan (Group A) Table III, subgroup 1 tests shall be performed on a production lot basis. Samples shall be selected in accordance with table IV. If one or more defects are found, the lot shall be screened for that particular defect and defects removed A new sample of parts shall be selected in accordance with table IV and all group A tests again performed If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE III. Group A Inspection. 1/

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Visual and mechanical inspection:			
Material	3.3	4.6 1	
Finish	3.3.1	4.6 1	
Dissimilar metals	3.3.2	4.6.1	
Design and construction	3.4	4.6.1	See table IV
Marking	3.23	4 6 1	
Workmanship	3.24	4.6.1	
Seal 2/	3.9	4.6.6	
Hermetic seal connectors	3.9.1	4 6.6.1	
Pressurized and waterproof	3.9.2	4.6.6.2	
Dielectric withstanding voltage	3.12	4.6.9	

1/ Verification may be accomplished using the manufacturer's process controls providing these controls are clearly equal to or more stringent than the requirements of this specification.

2/ These are in-process tests (100% inspection required).

4.5.1.2.2 Visual inspection (group A inspection). Each connector shall be visually examined for completeness, workmanship, and identification requirements. Attention shall be given to those assemblies that require a gasket to determine the condition of the gasket Gaskets missing, twisted, buckled, kinked, or damaged in any way shall be cause for rejection.

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TABLE IV Inspection level

Lot size	Visual and mechanical inspection	
	Major	Minor 1/
1 to 8	All	5
9 to 15	All	5
16 to 25	20	5
26 to 50	20	5
51 to 90	20	7
91 to 150	20	11
151 to 280	20	13
281 to 500	47	16
501 to 1,200	47	19
1,201 to 3,200	53	23
3,201 to 10,000	68	29
10,001 to 35,000	77	35
35,001 to 150,000	96	40
150,001 to 500,000	119	40
500,001 and over	143	40

1/ Samples may be pulled from either the production lot itself or from samples pulled from the lot for major defect testing

NOTES

1. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

2. Minor defect: A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

3. Critical defect: A critical defect is a defect that judgment and experience indicate is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product; or a defect that judgment and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table V in the order shown, and shall be made on sample units which have been subjected to and passed the group A inspection. Connectors having identical piece parts may be combined for lot purposes and shall be in proportion to the quantity of each PIN numbered connector produced.

4.5.1.3.1 Group B sampling plan. A sample of parts shall be randomly selected in accordance with table VI If one or more defects are found, the lot shall be screened for that particular defect and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected and subjected to all tests in accordance with table V. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification

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TABLE V Group B inspection

Inspection	Requirement paragraph	Test method paragraph
Force to engage/disengage	3.5	4.6.2
Coupling proof torque	3.6	4.6 3
Design and construction	3 4	4.6.1
Force to engage/disengage	3 5	4.6.2
Contact gaging	3 7	4.6.4
Permeability of nonmagnetic materials	3 8	4.6.5
Insulation resistance	3.10	4.6 7
VSWR 1/	3 14	4.6.11

1/ Destructive test, non deliverable

TABLE VI. Inspection level.

Lot size	Sample size	VSWR sample size
1 to 8	5	1
9 to 15	5	1
16 to 25	5	2
26 to 50	5	2
51 to 90	5	3
91 to 150	11	3
151 to 280	13	3
281 to 500	16	3
501 to 1,200	19	5
1,201 to 3,200	23	5
3,201 to 10,000	29	5
10,001 to 35,000	35	5
35,001 to 150,000	40	8
150,001 to 500,000	40	8
500,001 and over	40	8

4.5.1.3.2 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order, if the lot is accepted. Any connector deformed or otherwise damaged during testing shall not be delivered on the contract or order.

4.5.2 Periodic inspection. Periodic inspection shall consist of group C. Except where the results of these inspections shown noncompliance with the applicable requirements (see 4.5.2.1.4), delivery of products which have passed groups A and B shall not be delayed pending the results of these periodic inspections.

4.5.2.1 Group C inspection. Group C inspection shall consist of the inspections specified in table VII, in the order shown Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection.

4.5 2 1 1 Sampling plan. Six sample units of the same part number shall be selected from the first lot produced after the date of notification of qualification Thereafter, six sample units of the same part number shall be selected from current production after 200.000 connectors have been produced, or not less than once every year, whichever occurs first The sample units shall be divided equally and subjected to the inspections of the two subgroups

4 5 2 1 2 Failures. If one or more sample units fails to pass group C inspection, the lot shall be considered to have failed

4.5 2 1 3 Disposition of sample units Sample units which have been subjected to group C inspection

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shall not be delivered on the contract or order

TABLE VII Group C inspection 1/

Inspection	Requirement paragraph	Test method paragraph
Subgroup 1		
Center contact retention	3 11	4.6.8
Corrosion	3 13	4 6.10
Force to engage/disengage	3.6	4.6.2
Subgroup 2		
VSWR	3.14	4.6 11
RF transmission loss 1/	3.15	4.6.12
RF leakage 1/	3.16	4.6.13
Connector durability	3.17	4.6.14
Contact gaging	3.7	4.6 4
Force to engage/disengage	3.5	4 6.2
Subgroup 3		
Contact resistance	3.18	4 6.15
Thermal shock	3.19	4.6 16
Dielectric withstanding voltage	3.12	4.6.9
Contact resistance	3.11	4.6.8
VSWR	3.14	4.6.11
Moisture resistance	3.20	4.6.17
Dielectric withstanding voltage	3.12	4.6.9
Insulation resistance	3.10	4.6.7
RF high potential withstanding voltage 1/	3.21	4.6.18
Coupling mechanism retention force	3.22	4.6.19
Design and construction	3.4	4.6.1
Force to engage/disengage	3.5	4.6.2

1/ These test are only to be performed during initial qualification, as long as the qualifying design and manufacturing process does not change.

4.5.2.1.4 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subjected to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4 5.3 Inspection of packaging The sampling and the inspection of the preservation, packing and container marking shall be in accordance with the requirements of MIL-C-55330 "

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Review activities
Army - AR, AT, MI
Navy - AS, MC, SH
Air Force - 19, 99
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Preparing activity
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